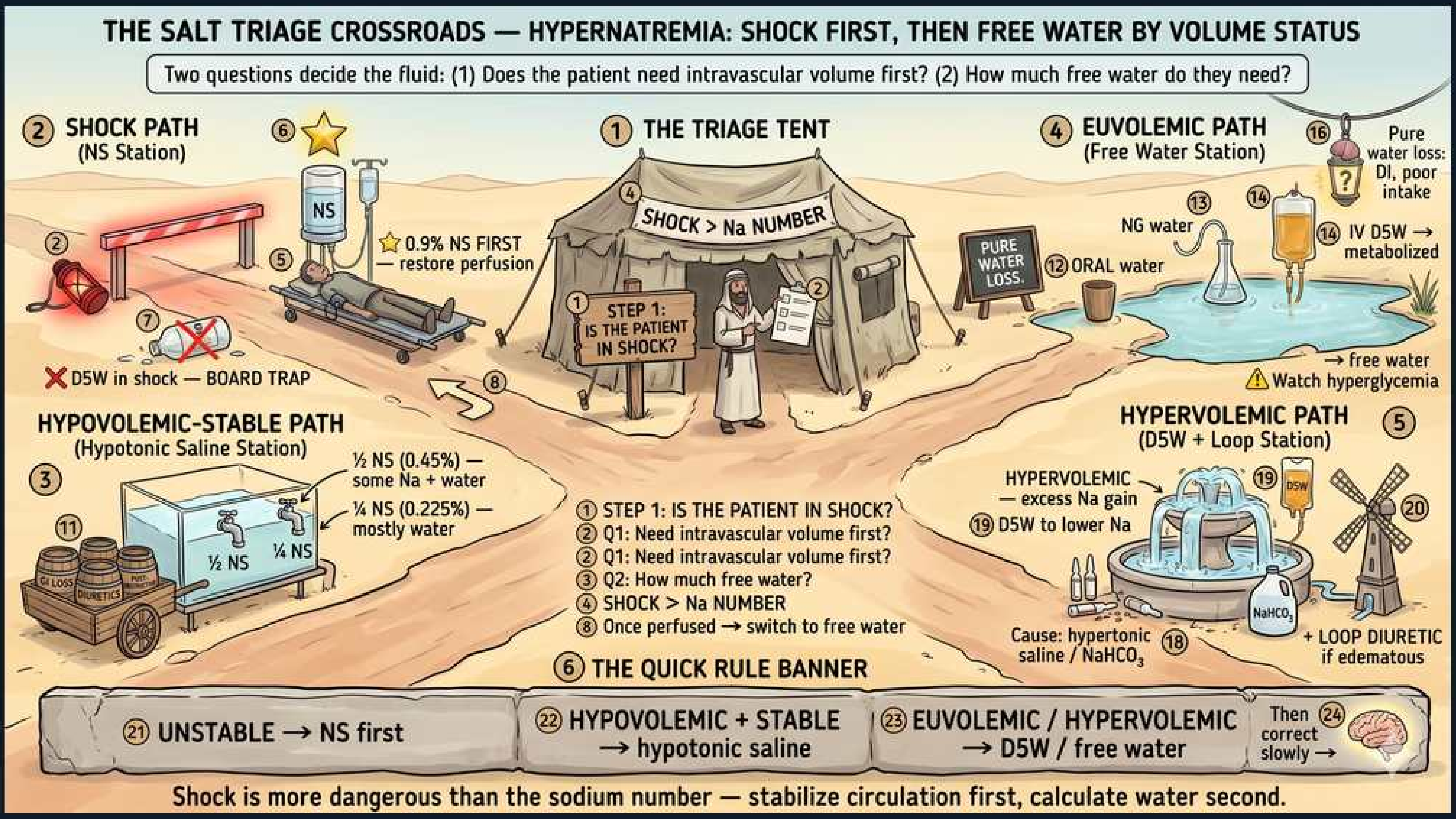
Hypernatremia — Fluid Choice Algorithm



1. Two questions

WHAT YOU SEE

Header: (1) Does the patient need IV volume first? (2) How much free water do they need?

WHAT IT ENCODES

Hypernatremia (Na >145 mEq/L) is always a free-water DEFICIT relative to sodium, but fluid choice is driven by two sequential questions. First: is the patient hemodynamically unstable and needing intravascular volume? Second: how large is the free-water deficit and how fast may it be corrected? Answering them in order prevents giving the wrong tonicity fluid.

BOARD TRAP

WRONG: choosing the corrective fluid (D5W) based purely on the sodium number before assessing volume status/shock. Discriminator — a hypotensive hypernatremic patient needs perfusion (isotonic saline) first; the Na number is corrected second, never the other way around.

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2. Step 1: in shock?

WHAT YOU SEE

Tent STEP 1: IS THE PATIENT IN SHOCK?

WHAT IT ENCODES

Step 1 is always hemodynamic: assess for shock/hypoperfusion (hypotension, tachycardia, oliguria, lactate, poor cap refill) before touching the sodium. Hemodynamics outrank the sodium number because circulatory collapse kills faster than hypertonicity. Only after circulation is secured do you move to the free-water-deficit math.

BOARD TRAP

WRONG: jumping straight to free-water deficit calculation in a clinically dry, hypotensive patient. Discriminator — restore perfusion first; deficit replacement is meaningless if the patient is in shock.

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3. Shock > Na number

WHAT YOU SEE

Sign SHOCK > Na NUMBER

WHAT IT ENCODES

Mnemonic anchor: 'Shock beats the Na number.' Circulatory stabilization always takes priority over correcting tonicity. A markedly elevated Na is dangerous, but hypoperfusion/hypovolemic shock is immediately lethal, so you stabilize the circulation first and correct sodium gradually afterward.

BOARD TRAP

WRONG: prioritizing rapid Na lowering over blood pressure in an unstable patient. Discriminator — perfusion first; sodium is a 48–72h problem, shock is a now problem.

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4. Shock → NS first

WHAT YOU SEE

NS STATION: 0.9% NS first, restore perfusion

WHAT IT ENCODES

If hemodynamically unstable, give 0.9% normal saline (isotonic) boluses first to refill the intravascular space and restore perfusion. Although NS contains 154 mEq/L Na (higher than plasma in absolute terms), it is still HYPOtonic relative to a patient whose serum Na is, say, 165, so it does not worsen hypernatremia while it resuscitates. Switch to free-water replacement only once the patient is perfused.

BOARD TRAP

WRONG: withholding isotonic saline because 'it has sodium' in a hypernatremic patient. Discriminator — NS is relatively hypotonic to severely hypernatremic serum and is the correct resuscitation fluid; perfusion is the priority.

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5. D5W in shock = trap

WHAT YOU SEE

Red X 'D5W in shock — BOARD TRAP'

WHAT IT ENCODES

D5W (free water once glucose is metabolized) distributes across total body water and only ~1/12 stays intravascular, so it is useless for resuscitation. Giving it to a shocked patient fails to restore perfusion and can drop glucose-driven osmoles abruptly. Resuscitate with isotonic crystalloid (NS), then correct the deficit with free water.

BOARD TRAP

WRONG ANSWER on the exam: 'give D5W' to resuscitate the hypotensive hypernatremic patient. Why wrong — D5W does not expand the intravascular volume; only ~8% stays in the vessels. Discriminator — use NS for perfusion, reserve D5W for deficit correction after stabilization.

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6. Switch to free water

WHAT YOU SEE

'Once perfused → switch to free water'

WHAT IT ENCODES

Once perfusion is restored and the patient is no longer in shock, switch from isotonic saline to hypotonic/free water (½ NS, ¼ NS, D5W, or oral water) to actually correct the elevated sodium. This two-phase approach — resuscitate, then correct — is the core algorithm. Continuing NS after perfusion is restored will fail to bring the Na down.

BOARD TRAP

WRONG: staying on 0.9% NS after the patient is resuscitated. Discriminator — NS maintains volume but won't correct hypernatremia; you must transition to free water to lower the Na ≤10 mEq/L/day.

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7. Hypovolemic-stable

WHAT YOU SEE

HYPOTONIC SALINE STATION: ½ NS (0.45%) or ¼ NS (0.225%)

WHAT IT ENCODES

Hypovolemic but hemodynamically stable (e.g. osmotic diuresis, partial volume loss): use hypotonic saline (½ NS 0.45% or ¼ NS 0.225%) which simultaneously replaces some volume AND delivers free water to correct Na. This straddles the two needs — unlike NS (volume only) or D5W (free water only).

BOARD TRAP

WRONG: giving pure D5W to a still-volume-depleted patient. Discriminator — if the patient is hypovolemic (not euvolemic), they need some sodium with their water, so hypotonic saline, not D5W, is the answer.

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8. ½ NS vs ¼ NS

WHAT YOU SEE

½ NS = some Na + water; ¼ NS = mostly water

WHAT IT ENCODES

½ NS (0.45%, 77 mEq/L Na) provides moderate volume plus free water — roughly half of each liter acts as free water. ¼ NS (0.225%, ~38 mEq/L Na) is mostly free water with minimal sodium, for milder volume deficits with a larger free-water gap. Choose based on how much volume vs free water the patient needs.

BOARD TRAP

WRONG: treating ½ NS and ¼ NS as interchangeable. Discriminator — ½ NS for greater volume need, ¼ NS when the volume deficit is small but the free-water deficit is large.

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9. Euvolemic → D5W/oral

WHAT YOU SEE

FREE WATER STATION: oral water, D5W, pure water loss = DI

WHAT IT ENCODES

Euvolemic hypernatremia = PURE water loss without significant Na loss (diabetes insipidus, insensible/respiratory losses, restricted access to water in elderly/intubated). Give pure free water: oral water is preferred whenever the gut works and the patient can drink/take NG water, otherwise IV D5W. Address the cause too (e.g. desmopressin for central DI).

BOARD TRAP

WRONG: routinely starting IV D5W when the patient could simply drink water. Discriminator — oral/enteral free water is the safest, cheapest first choice in a euvolemic patient with an intact, accessible gut; IV D5W is reserved for those who can't take enteral water.

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10. Watch hyperglycemia

WHAT YOU SEE

Warning: D5W metabolized → watch hyperglycemia

WHAT IT ENCODES

D5W delivers free water only after the dextrose is metabolized; rapid infusion can outpace glucose clearance, causing hyperglycemia and a resulting osmotic diuresis that paradoxically worsens free-water loss. Monitor glucose, keep infusion rates measured (often ≤ a few hundred mL/h), and check Na every 2–4h during active correction.

BOARD TRAP

WRONG: running D5W fast to 'catch up' the deficit. Discriminator — fast D5W → hyperglycemia → osmotic diuresis → more water loss AND risks overcorrection; correct slowly and monitor glucose and Na.

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11. Hypervolemic → D5W + loop

WHAT YOU SEE

D5W + LOOP STATION: excess Na gain, D5W to lower Na, loop diuretic if edematous

WHAT IT ENCODES

Hypervolemic hypernatremia = sodium GAIN, not water loss (hypertonic saline infusion, sodium bicarbonate during codes, excess salt, mineralocorticoid excess). The problem is too much total-body sodium, so management is D5W to add free water PLUS a loop diuretic (furosemide) to excrete the excess sodium; consider dialysis if renal failure.

BOARD TRAP

WRONG ANSWER: giving isotonic or hypotonic SALINE to a hypervolemic (Na-overload) hypernatremic patient. Why wrong — you'd be adding more sodium to a sodium-overloaded, edematous patient. Discriminator — Na-gain states need sodium REMOVAL (loop diuretic ± D5W), the opposite of the volume-depleted pathways.

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12. Unstable → NS first

WHAT YOU SEE

Banner: UNSTABLE → NS first

WHAT IT ENCODES

Quick-rule banner #1: UNSTABLE = 0.9% NS first. Isotonic saline restores perfusion regardless of the sodium number; you correct the Na after the hemodynamics are stabilized. This is the single most tested branch of the algorithm.

BOARD TRAP

WRONG: choosing free water/D5W for the unstable patient. Discriminator — instability always routes to isotonic saline first.

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13. Hypovol+stable → hypotonic

WHAT YOU SEE

Banner: HYPOVOLEMIC + STABLE → hypotonic saline

WHAT IT ENCODES

Quick-rule banner #2: HYPOVOLEMIC + STABLE = hypotonic saline (½ NS or ¼ NS) because the patient needs both some volume and free water. This is the middle path between pure resuscitation (NS) and pure free-water correction (D5W).

BOARD TRAP

WRONG: giving pure D5W (free water only) to a hypovolemic patient who still needs some volume. Discriminator — stable + hypovolemic → hypotonic saline, not D5W.

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14. Euvolemic → D5W/free water

WHAT YOU SEE

Banner: EUVOLEMIC → D5W / free water; then correct slowly

WHAT IT ENCODES

Quick-rule banner #3: EUVOLEMIC = D5W / oral free water (pure water replacement) AND always correct slowly — lower Na ≤10 mEq/L per 24h. The slow-correction caveat applies to every pathway: too-fast lowering of a chronically high Na causes cerebral edema and seizures.

BOARD TRAP

WRONG: correcting the euvolemic patient's Na quickly toward 140 in one day. Discriminator — regardless of pathway the maximum safe drop is ~10 mEq/L/24h; faster correction risks fatal cerebral edema.
